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Machine Setup and Component Placement in Printed Circuit Board Assembly

JONATHAN F. BARD
Professor, Graduate Program in Operations Research and Industrial Engineering, Department of Mechanical Engineering, University of Texas, Austin, TX 78712-1063

RAYMOND W. CLAYTON
Systems Analyst, Optimization Alternatives, Austin, TX 78703

THOMAS A. FEO
Associate Professor, Graduate Program in Operations Research and Industrial Engineering, Department of Mechanical Engineering, University of Texas, Austin, TX 78712-1063

Abstract. Populating printed circuit boards is one of the most costly and time-consuming steps in electronics assembly. At the beginning of each work order, three decisions are required: (1) a sequence must be specified for placing the individual components on the board; (2) tape reels must be assigned to positions on the magazine rack; and (3) a retrieval plan must be determined should the same component type be assigned to more than one magazine slot. Collectively, these problems can be modeled as a nonlinear integer program. In this paper, we develop a series of algorithms for solving each using an iterative two step approach.

Initially, a placement sequence is generated with a weighted, nearest neighbor traveling salesman problem (TSP) heuristic; the two remaining problems are then formulated as a quadratic integer program and solved with a Lagrangian relaxation scheme. As a final step, the current magazine assignments are used to update the placement sequence, and the entire process is repeated.

Our ability to deal, at least in part, with simultaneous machine operations represents the major contribution of this work. The methodology was simulated for a set of boards obtained from Texas Instruments and theoretically compared with a heuristic currently in use.

Key Words: printed circuit board assembly, surface mount technology, quadratic integer programming, dynamic programming.

1. Introduction

As product lifetimes decrease and customer demands become less predictable, an increasing number of manufacturers are turning to flexible automation to cut costs. This is most evident in the electronics industry where the dominant activity has become printed circuit board (PCB) assembly. Because of the large number of components that must be placed on a single board, and the precision required at each step of the process, automation is an economic necessity.

Marcoux (1986) reports that over 28% of the final cost of a PCB is directly attributable to assembly, the bulk of which involves component placement. The machines that perform these operations are at the forefront of technology, and are designed with both flexibility and speed in mind. For the most part, they must be reprogrammed for each work order, implying that production efficiency is closely tied to the setup and sequencing decisions.

The purpose of this paper is to examine the problems surrounding PCB assembly and to develop a set of algorithms that can be used to aid shop floor personnel better plan their runs. The focus is on the placement of surface mount components with a multihead machine. Our interest is in high-volume manufacturing with the objective of reducing component placement times for individual batches.

The next section presents some background material and describes the machine being modeled. In sections 3 and 4, the problem is formulated as a quadratic integer program and a solution methodology involving Lagrangian relaxation is discussed. Data requirements and the parameters associated with the boards used in the study are identified in section 5. In section 6, we examine issues related to implementation and present our computational results. We investigate 12 different board types in production at Texas Instruments' (TI) Austin facility and theoretically compare our algorithm with an improved version of a heuristic recently put into practice. We close with a critique of the proposed methodology and suggest some topics for future research.

2. Surface mount technology and PCB assembly

In recent years, surface mount technology (SMT) has begun to replace the standard pin-through-hole design of electronic components. Whereas the latter requires mechanical and electrical connections to be made by extending wire leads through the board, SMT permits the leads to simply sit on the surface. The benefits resulting from this innovation have been significant. Bowlby (1985) cites an example where a 69% space savings was achieved with a typical 68 lead surface mount chip compared to the traditional dual-in-line package (DIP). The smaller packages allow a much larger number of pins on a single component. In addition, because there are no leads extending through the board, components may be placed on either side with proportionate benefits.

Along with size and cost savings, SMT offers advantages to both circuit designers and production engineers. Because the components sit on the surface, there is no need for complex finger manipulations to position the leads as in pin-through-hold insertion. Nevertheless, component placement must be increasingly accurate due to smaller electrical connection pads and denser designs.

2.1. Problem description

In the analysis, we focus on the Fuji CP II, which is one of the most sophisticated pick-and-place machines on the market. Its main features are a stereoscopic vision system for inspection, a placement rate of over 12,000 components per hour, and an error rate of less than 1 in 10^4. Other machines such as the Panasert MKI-LL possess many of these same characteristics [see Grotzinger and Sciomachen (1988) for more detail].

The Fuji CP II is known in the industry as a "chip shooter." At the TI Austin plant, it is used to place small resistors, capacitors, and transistors, and is often the source of the bottleneck. Large components such as quad flat packs and plastic leaded chip carriers (PLCCs) are placed by a second machine in series.

A schematic of the Fuji CP II highlighting the PCB table, the magazine rack, and the placement head carousel is presented in figure 1. During assembly the board lies on a movable table in the center of the machine having two axes of motion (X and Y). For each component to be attached, the table moves the board so that its (X, Y) position is directly under the fixed placement station of the carousel. A magazine rack containing up to 100 positions for different component types is located on the near side of the machine. (Although the CP II only accepts tape reels, future models are expected to permit the use of vibratory slope feeders as well.) The rack moves back and forth on a single axis in order to bring a reel with the required component type in line with the pickup station. Above the PCB table and adjacent to the magazine rack is a carousel containing 12 placement heads. Each head has a pair of nozzles that grip and hold the components until they are placed on the board. The two nozzles are of different diameter, and switch automatically depending on the size of the component to be retrieved. Only one nozzle is used at a time. After a head picks up a component from the magazine (gripping), the carousel rotates clockwise to the placement station so the component can be seated on the board. The machine must be programmed so that each head receives a component from the correct magazine slot and the PCB is in the proper position for the component to be placed.

As the carousel rotates, several additional operations are performed between the gripping and placement stations. Each component is examined by an automated visual inspection system for orientation and diagnostics. If an uncorrectable problem is identified, the component is ejected at the following station and reassigned to the end of the sequence. Otherwise, the component is oriented for placement. If any mishaps occur at this stage, a second ejection station is provided. Finally, there are two stations to set up and reorient the placement nozzle to grip the next component. These additional operations occur in parallel with

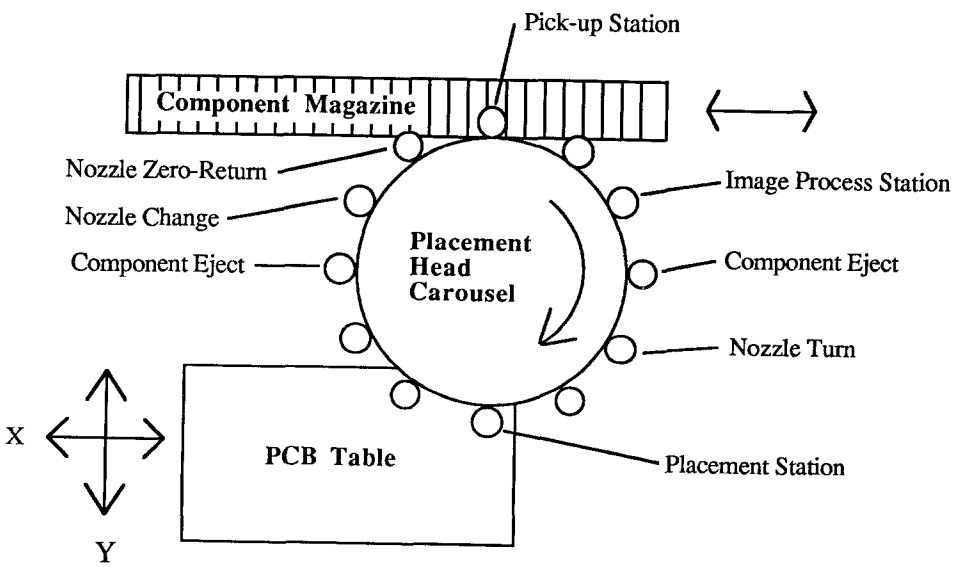


Figure 1. Schematic of Fuji CP II.

and are dominated by the gripping and placement operations and thus, do not constrain the timeline.

The PCB assembly process can be broken down into three subproblems. The first involves the specification of the component placement sequence which determines the movement of the table (*table tour*). The second is the assignment of component types to magazine positions (*component assignment*). The third is concerned with the retrieval of components (*component retrieval sequence*) and determines the movement of the magazine rack should more than one slot be assigned the same component type.

A closer examination of these subproblems reveals that each by themselves may be difficult to solve. Consequently, we take a hierarchical approach and first develop a good *table tour* using a weighted, nearest neighbor traveling salesman (TSP) heuristic [related work is reported by Danusaputro et al. (1989), and Leipala and Nevalainen (1989)]. Next, the *component assignment* and *component retrieval* problems are solved jointly with a Lagrangian relaxation scheme for the given placement sequence. The algorithm developed for this purpose represents the major contribution of this work. Finally, we follow the common practice of redetermining the table tour after the magazine assignments have been made, and then resolve the latter two problems in an attempt to find a locally better solution. This step introduces an element of feedback in the methodology so it should not be thought of as strictly hierarchical.

2.2. *Placement operation timeline*

Before any of the above problems can be formulated, it is necessary to gain an understanding of machine operations. The parameters of interest are the table movement rates (X and Y), the magazine movement rate, the placement head carousel rotation rate, and the gripping and placement operation times. Nominal data for the Fuji are displayed in table 1.

Figure 2 shows a typical timeline for machine operations. The vertical lines point to either the completion of one operation or the beginning of another. In general, the table may start to move as soon as the previous component is placed. Similarly the magazine movement may begin as soon as the previous component's gripping action is complete. Carousel rotation may not start until both the placement and gripping operations are finished.

For purposes of illustration, figure 2 shows the machine in an arbitrary state and assumes that all nozzles are occupied. At time 0 the carousel rotation has just begun, the table is moving to the ith placement location, and the magazine is traveling to retrieve the component for the $i + 6$th placement. At time 3 the carousel rotation is complete, and since

Table 1. FUJI CP II operating rates.

Parameter	Value
Table movement rate (X and Y)	20 mm/0.15 seconds
Magazine movement rate	1 slot/0.15 seconds
Carousel rotation rate	1 station/0.15 seconds
Gripping action	0.10 seconds
Placement action	0.10 seconds

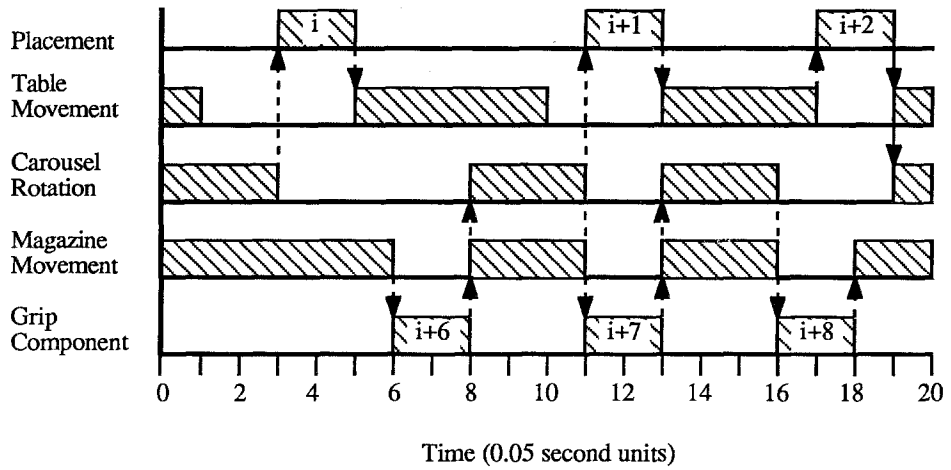


Figure 2. Example of timeline for placement operations.

the table movement finished earlier the placement operation may begin. At time 5 the placement is complete and the table may begin to move to the i + 1st placement location. However, the magazine movement does not finish until time 6, and the subsequent retrieval of the i + 6th component is completed at time 8. The next carousel rotation begins at time 8 when both the placement of component i and the retrieval (gripping) of component i + 6 are finished. At time 11 the carousel rotation is complete. Since the table movement is finished at time 10, component placement starts immediately. Similarly, the magazine movement for this case is finished by time 11, so the retrieval of component i + 7 may start.

It is important to note the minimum time required for each placement cycle. If there are no delays due to magazine movement and table movement, then the placement sequence time is fixed. Specifically, for the parameters in table 1, any magazine or table movements that are executed in 0.15 seconds or less (the time of a carousel rotation) are equally good. This factor comes into play when we consider a *zero penalty placement sequence* in the development of our table tour heuristic. In particular, we exploit the fact that any location selected within the zero penalty movement distance (20 mm in either direction) does not induce additional table tour time. This property was originally noted by Ahmadi et al. (1988) and Grotzinger (1988), where it was referred to as "free" movement.

2.3. Related literature

Very few analytic papers have been published on PCB assembly. Ball and Magazine (1988) were the first to discuss the problem of optimizing the placement sequence (*table tour*) given the magazine assignments. Their methodology includes the use of a heuristic that guarantees optimality if the gripper movement is rectilinear, a restriction that accompanied earlier technology. When this is not the case, they provide a bound on the difference between their results and the optimum. The possibility of assigning the same component to multiple magazine positions is not considered.

Ahmadi et al. (1988, 1991) present a method for optimizing component allocation to magazines for the Dyna-Pert MPS 500 dual delivery placement machine. They point out that the key to success is to take advantage of concurrent operations as much as possible. Their method attempts to partition assignments of component reels to balance the retrieval operations while minimizing the need to change nozzles. They also discuss allocation of components to magazine slots to maximize the number of boards to be completed before components must be restocked. They take the view that setup times are significant due to extensive fine-tuning of the feeders, and that component assignments should be made to reflect a batch environment with many board types. (Since their study was completed, notable advances in technology have lessened the importance of setup times.) An extension of this work using a Petri net-based simulation language is given by Archetti et al. (1987); also see Grotzinger and Sciomachen (1988).

Other authors have examined related issues. In particular, Randhawa et al. (1985) developed an integer programming model to determine optimum decisions for capital investment in component placement sequencing equipment given set production requirements. The model minimizes a combination of capital, operating, and maintenance costs. They also discuss the problem of job scheduling on component sequencing machines to minimize preproduction time, and reference TSP and other heuristics to solve this problem.

Lofgren and McGinnis (1986a) consider PCB assembly when many different board types are to be produced. They first sequence the boards and then, given the sequence, optimize setup. The goal is to increase efficiency by minimizing the machine setup time over the entire production period. In a related paper, Lofgren and McGinnis (1986b) discuss configuration problems in electronics manufacturing. Here they develop algorithms to determine the set of operations that each placement machine will perform given that more than one option is available in a work cell. The goal is to minimize makespan by effectively balancing the workload between the placement machines without violating system constraints.

3. Problem formulation

In this section, we give a mathematical formulation of the *component assignment* and *component retrieval* problems. The TSP heuristic used for the *table tour* is outlined in Appendix B. For the PCBs under consideration, all tape reels are of equal size. The following notation is used in the development:

I = ordered set of locations (X, Y) on the PCB (obtained from TSP heuristic)
n_I = number of placement locations, $n_I = |I|$
M = set of magazine positions
n_M = number of slots on the magazine rack
P = set of unique components
n_P = number of unique component used on the PCB
i = index of placement locations on the PCB, $i = 1, \ldots, n_I$
p = index of component types, $p = 1, \ldots, n_P$
j, k = indices for magazine locations, $j, k = 1, \ldots, n_M$
x_{ij} = binary decision variable equal to 1 if the component type associated with the ith placement is in magazine position j, 0 otherwise

y_{jp} = binary decision variable equal to 1 if component type p is in magazine position j, 0 otherwise
a_{ip} = constant equal to 1 if placement location i requires component type p, 0 otherwise
η_p = maximum number of magazine positions component type p may occupy
$t_{jk}(i-1, i)$ = incremental amount of time required to complete the retrieval of the ith component from position k on the magazine rack given that the $i-1$st component was just retrieved from magazine position j
I_p = set of placement locations that require component type p and
n_D = number of placement head positions between the placement station and the gripping station (for the Fuji CP II, $n_D = 6$).

In the formulation, it will be assumed that when processing begins, the placement head carousel is empty and that the magazine rack and table are in the proper positions for retrieving and placing the first component. It should be noted that the Fuji is programmed to complete all operations on one board before it begins another. Therefore, it is not possible to overlap the retrieval of the first n_D components for the next board while placing the last few components on the current board. In addition, we assume that no head in the carousel is left empty and that ejection rates during inspection are negligible. The high reliability of the Fuji CP II coupled with the newly implemented total quality control program at TI is the basis for the latter assumption [overall system performance is discussed in Feo et al. (1990)]. If a component is rejected, the current practice is to recyle the board after the initial pass. Other options may be considered, however. One possibility is to reinsert the component in the remaining sequence so that it occasions the smallest delay in completion time.

With these factors in mind, the component assignment and component retrieval problem is given by:

$$F^* = \min f(x, y) = \sum_{i=2}^{n_I} \sum_{j=1}^{n_M} \sum_{k=1}^{n_M} t_{jk}(i-1, i) x_{i-1,j} x_{ik} + \tau \tag{1a}$$

subject to

$$\sum_{j=1}^{n_M} x_{ij} = 1 \qquad i = 1, \ldots, n_I \tag{1b}$$

$$\sum_{i=1}^{n_I} a_{ip} x_{ij} \leq |I_p| y_{jp} \qquad j = 1, \ldots, n_M;\ p = 1, \ldots, n_P \tag{1c}$$

$$\sum_{j=1}^{n_M} y_{jp} \leq \eta_p \qquad p = 1, \ldots, n_p \tag{1d}$$

$$\sum_{p=1}^{n_P} y_{jp} \leq 1 \qquad j = 1, \ldots, n_M \tag{1e}$$

$$x_{ij} = 0, 1;\ y_{jp} = 0, 1 \qquad \forall\ i, j, p \tag{1f}$$

The objective in (1a) is to minimize the total time it takes to grip and place all the components on the board. For a given sequence, I, this is found by summing the time between consecutive component retrievals, and then adding some fixed time, τ, for placing the final n_D components.

The first constraint (1b) serves to ensure that each location on the PCB receives a component from exactly one magazine position. Similarly, (1c) permits components to be retrieved from only magazine positions that have been assigned the required component type, while (1d) limits the number of magazine positions to which a component type may be assigned. These constraints are similar to those used in modeling tool setups on CNC machines (Berrada and Stecke 1986; Hankins and Rovito 1984; Kim and Yano 1992; Stecke 1983).

In practice, it is common to "stationalize" (prefix) assignments of high-volume components to avoid frequent changeovers. Beyond this, several factors must be taken into account when specifying η_p. For example, the manufacturer may want to limit the number of reels for those components that are expensive and in less demand. Also, it may be necessary to impose this restriction when only a few reels of a certain component type are available, or when inventories are low and replenishment delays are expected.

The final set of structural constraints (1e) guarantees that each position on the magazine rack is allocated no more than one component type. As stated, it is assumed that each reel occupies exactly one slot on the magazine. While this is the way TI sets up up the CP II, other manufacturers might find it desirable to use different size reels on the same machine. This can be taken into account at the expense of greatly complicating the model by adding a series of constraints in the y variables to exclude the use of neighboring slots occupied by double-sized reels [see Grotzinger (1988)]. Our approach, as detailed in this paper, is only valid for reels that fit into a single slot.

4. Solution methodology

Problem (1) is a quadratic integer program (IP) which, at best, is difficult to solve with standard techniques for even unrealistically small problems (Liggett 1981). The difficulty is compounded by the fact that the term $t_{jk}(i-1, i)$ in the objective function is dependent upon the yet to be determined component retrieval sequence, x, and magazine assignments, y. Because of concurrency of movement, the values of $t_{ji}(i-1, i)$ can only be found by examining *all* feasible combinations of x and y. This, of course, is equivalent to solving (1) by complete enumeration.

As an alternative, we propose to decompose (1) into two relatively easy to solve subproblems, and combine the results to obtain good solutions. The decomposition is achieved by relaxing the complicating constraint (1c). Branch and bound is used to reach optimality.

4.1. Problem relaxation

Let λ_{jp} be a fixed nonnegative multiplier associated with the corresponding inequality in (1c). If these constraints are "dualized" by placing them in the objective function (1a) as penalty terms, we get the following *relaxed primal problem*.

$$\theta(\lambda) = \min \left[\sum_{i=2}^{n_I} \sum_{j=1}^{n_M} \sum_{k=1}^{n_M} t_{jk}(i-1, i) x_{i-1,j} x_{ik} + \sum_{p=1}^{n_P} \sum_{j=1}^{n_M} \lambda_{jp} \sum_{i=1}^{n_I} a_{ip} x_{ij} - \sum_{j=1}^{n_M} \sum_{p=1}^{n_P} \lambda_{jp} |I_p| y_{jp} \right] \tag{2}$$

subject to (1b), (1d), (1e), (1f).

We are interested in the value of λ that maximizes $\theta(\lambda)$ because the solution to the *dual problem*

$$\theta^* = \max_{\lambda \geq 0} \theta(\lambda) \tag{3}$$

gives the tightest lower bound on the solution to (1).

The advantage of the above relaxation is that for λ fixed, (2) decomposes into two subproblems: a quadratic integer program in the x variables, and a transportation problem in the y variables.

Quadratic IP

$$\theta_1(\lambda) = \min_x \left[\sum_{i=2}^{n_I} \sum_{j=1}^{n_M} \sum_{k=1}^{n_M} t_{jk}(i-1, i) x_{i-1,j} x_{ik} + \sum_{p=1}^{n_P} \sum_{j=1}^{n_M} \lambda_{jp} \sum_{i=1}^{n_I} a_{ip} x_{ij} \right] \tag{4a}$$

subject to

$$\sum_{j=1}^{n_M} x_{ij} = 1 \qquad i = 1, \ldots, n_I \tag{4b}$$

$$x_{ij} = 0, 1 \qquad i = 1, \ldots, n_I; j = 1, \ldots, n_M \tag{4c}$$

Transportation problem

$$\theta_2(\lambda) = \min_y - \sum_{j=1}^{n_M} \sum_{p=1}^{n_P} \lambda_{jp} |I_p| y_{jp} \tag{5a}$$

subject to

$$\sum_{j=1}^{n_M} y_{jp} \leq \eta_p \qquad p = 1, \ldots, n_p \tag{5b}$$

$$\sum_{p=1}^{n_P} y_{jp} \leq 1 \qquad j = 1, \ldots, n_M \tag{5c}$$

$$\sum_{j=1}^{n_M} y_{jp} \geq 1 \qquad p = 1, \ldots, n_P \tag{5d}$$

$$y_{jp} = 0, 1 \qquad j = 1, \ldots, n_M; \; p = 1, \ldots, n_I \tag{5e}$$

where (5d) has been added to the original constraint set to assure that each component type is placed in at least one position on the magazine rack. This addition is permissible because every feasible solution to the original problem is also feasible to (5). Thus, the bounding properties of the relaxation are preserved.

Unfortunately, problem (4) still cannot be solved with a standard code because the objective function coefficients associated with the quadratic terms remain a function of the decision variables. However, if the ith component is known to be in the kth magazine position, it is an easy matter to determine the best position for the $i-1$st component. This observation is exploited in our dynamic programming approach discussed below.

Before presenting the algorithm, we state a result that foreshadows the presence of a duality gap, $F^* - \theta^*$. Sufficient conditions for this gap to be zero, as well as for the solution to the relaxed problem (2) to be optimal to (1) have been characterized by Geoffrion (1974). We derive necessary conditions for $F^* = \theta^*$. In so doing, we show that the difference between the objective function (1a) evaluated at any feasible point, and the best lower bound achievable, is determined by the amount of additional time imposed by the component retrieval decisions above the zero penalty time (0.15 seconds) for the given placement sequence. Proofs are contained in Appendix A.

Lemma 1. The optimal solution to the dual problem $\max[\theta(\lambda) : \lambda \geq 0]$ is $\lambda^* = 0$.

Lemma 2. Let (x^*, y^*) solve (1) and λ^* solve (3). A necessary condition for the absence of a duality gap is that the component retrieval plan associated with (x^*, y^*) does not impose any additional time on the placement operation beyond the zero penalty time. That is, given (x^*, y^*), the time it takes to place each component is restricted solely by the table tour and not the movement of the magazine rack.

Lemma 1 states that the dual problem (3) has a trivial solution; Lemma 2 states that for a duality gap not to arise, the optimal component retrieval plan must satisfy certain constraints that are not required to hold in a general instance of the problem. Taken together, the implication is that solving problem (2) is not in general sufficient to solve problem (1). An implicit enumeration scheme such as branch and bound is likely to be needed to guarantee optimality. In any case, it is worth noting that even if no duality gap exists, there is no assurance that the optimal solution to (2) with multipliers fixed at λ^* would in fact yield (x^*, y^*).

4.2. *General approach*

In view of Lemma 1, the main purpose of solving the relaxed problem (2) is not to solve the dual (3), but to generate good candidate magazine assignments, y, that can be used to construct feasible retrieval sequences, x. A procedure for doing this follows.

Step 1. Initialize the Lagrange multipliers, $\lambda^0 = 0$, and set incumbent $\bar{F} = \infty$.
Step 2. For λ^κ fixed at iteration κ, solve the relaxed problems (4) and (5) to obtain (x^κ, y^κ).
Step 3. For y^κ fixed, find a feasible solution to the retrieval subproblem to get $\bar{x}^k$ and $f(\bar{x}^\kappa, y^\kappa)$.
Step 4. Update the incumbent by setting $\bar{F} = \min[\bar{F}, f(\bar{x}^\kappa, y^\kappa)]$, and store the corresponding decision variables.
Step 5. Check stopping criterion.
Step 6. Update the multipliers, λ^κ, and go to Step 2.

The burdensome computations in this iterative scheme are performed at Step 2 in solving the quadratic IP and at Step 3 in solving the retrieval subproblem. The reason for solving problem (4) at each iteration is to obtain the information needed to update the mulitpliers at Step 6. A standard subgradient procedure is used for this purpose (Fisher 1981). In particular, let $\bar{F}$ be an upper bound on the primal objective function, F, and let γ_κ be a scalar satisfying $0 < \gamma_\kappa \leq 2$. The equation used to update λ is:

$$\lambda_{jp}^{\kappa+1} = \max\left\{\left[\lambda_{jp}^{\kappa} + s_\kappa\left(\sum_{i=1}^{n_I} a_{ip}x_{ij} - |I_p|\, y_{jp}\right)\right], 0\right\},$$

where the step size $s_\kappa = \dfrac{\gamma_\kappa[\bar{F} - \theta(\lambda^\kappa)]}{\Sigma_j\Sigma_p(\Sigma_i a_{ip}x_{ij} - |I_p|\, y_{jp})^2}$.

In the computations, the upper bound $\bar{F}$ is initially set to ∞ at Step 1 and iteratively updated at Step 4; γ_κ is fixed at 1. The stopping criterion used at Step 5 is a prespecified number of iterations (100 in the implementation). Other criteria are possible.

In essence, the above algorithm provides a systematic way of exploring the feasible region. Only in those instances, however, where the conditions of Lemma 2 are met and a feasible pair (x^*, y^*) yielding a zero duality gap is found, can the optimality of (1) be verified without additional work.

4.3. *Solving the quadratic IP*

At Step 2, the quadratic IP (4) must be solved to optimality. We now show how this can be done with a shortest-path algorithm. To begin, we define a graph $G(V, E)$ in the form of a layered network. The node set V of G consists of each placement location on the PCB duplicated once for every magazine retrieval position; the edge set E is formed by connecting each node in V to its immediate successors which correspond to the next placement location. Arc weights are given by

$$c_{jk}(i-1,i) = t_{jk}(i-1,i) + \sum_{p=1}^{n_p} \lambda_{kp} a_{ip} \tag{6}$$

where the second term is constant for i and k fixed. As such, these weights can be computed iteratively as the graph is traversed in the forward direction. Figure 3 shows a portion of the layered network. The shortest path through this network corresponds to an optimal retrieval sequence for the current values of λ. We propose to use dynamic programming (DP) to solve this problem.

For stage i and state k, the basic recursion is

$$f_i(k) = \min_{j \in M} \left[t_{jk}(i-1,i) + \sum_{p=1}^{n_p} \lambda_{kp} a_{ip} + f_{i-1}(j) \right]; \quad i = 1, \ldots, n_I; \; k = 1, \ldots, n_M \tag{7a}$$

$$f_0(-) = 0; \qquad t_{jk}(0, 1) = \Delta g = 0.1 \tag{7b}$$

where $f_i(k)$ is the minimum time to grip the first i components (and necessarily place the first $i-6$ components) given that the ith component is in magazine position k. The computation and memory requirements for (7) are of order $O(n_I n_M^2)$ and $O(n_I n_M)$, respectively.

From (7a), it can be seen that the time it takes to complete the retrieval of the first i components is totally dependent on the decisions made for the ith component and the state of the system at the $i-1$st stage. Thus, the forward DP approach will solve the problem by stepping through the placement sequence making decisions for the next stage based on the current assignments and the results from the previous placement. This allows us to calculate $t_{jk}(i-1,i)$ recursively. Some additional notation is needed to show how this

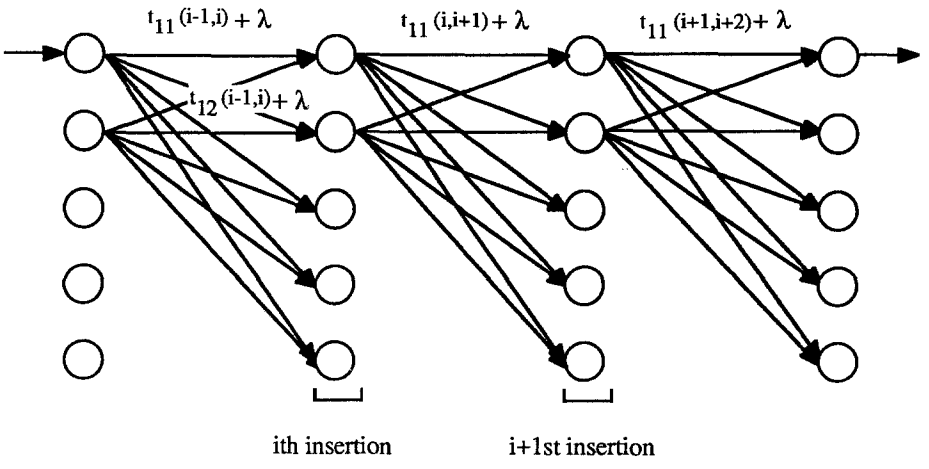


Figure 3. Layered network for solving the quadratic IP.

is done. In the presentation, it is assumed that operations begin at "time" zero. All absolute references to time (such as "time placement begins") are made with respect to zero. Additional notation used is:

tp_i = the time placement begins for ith component
Δp = incremental time to place a component and clear the board (for the Fuji, $\Delta p = 0.1$ seconds)
tg_i = the time pickup head begins to grip ith component, extraction of component from feeder
Δg = incremental time for head to grip and remove component from magazine ($\Delta g = 0.1$ seconds)
tr_i = the time carousel begins to rotate after having picked up the component for the ith placement
Δr = incremental time for the gripper to move from pickup position to the next station ($\Delta r = 0.15$ seconds)
tt_i = time table movement begins to set up for ith component placement
$\Delta t_{i-1,i}$ = incremental time that is required for the table to move to the ith placement location from the previous location
tm_i = time magazine movement begins for pickup of ith component and
$\Delta m_{i-1,i}$ = incremental time that is required to move the magazine rack for pickup of the ith component from the previous magazine position.

The time between successive component grippings, $t_{jk}(i-1, i)$, can now be obtained from these definitions:

$$t_{jk}(i-1, i) = tg_i - tg_{i-1}. \tag{8}$$

In order to evaluate this expression, it is first necessary to calculate the other time components. The recursive equations used for this purpose are discussed below. A basic assumption in the development is that no head in the placement carousel is left empty between the gripper and placement stations once processing starts.

Component placement: tp_i. The placement of component i occurs as soon as the table and carousel have completed their appropriate movements. For all but the first component, the table must move from location $i-1$ to the current placement location. The carousel must rotate so that component i is at station $n_D + 1$; this corresponds to the movement of component $i + (n_D - 1)$ from the gripping position to the next station. The first component may be placed as soon as it reaches the placement station which occurs when the head rotation for component n_D has been completed.

$$tp_i = \max(tr_{i+(n_D-1)} + \Delta r,\ tt_i + \Delta t_{i-1,i}), \qquad i = 2, \ldots, n_I$$

$$tp_1 = tr_{n_D} + \Delta r.$$

Gripper movement (component extraction): tg_i. The gripping operation for component i occurs when the carousel rotation for the previous component is complete and the magazine rack has moved into the correct position for the pickup. Gripping of the first component occurs at time 0 since the magazine is assumed to be in the proper position.

$$tg_i = \max(tr_{i-1} + \Delta r,\ tm_i + \Delta m_{i-1,i}) \qquad i = 2, \ldots, n_I$$

$$tg_1 = 0$$

Placement head rotation: tr_i. It takes n_D (=6) rotations to fill up the carousel before the first component is placed. For the first n_D movements, the head may rotate as soon as the component gripping is complete.

$$tr_i = tg_i + \Delta g. \qquad i = 1, \ldots, n_D$$

When the carousel is full, it may rotate only after the gripping operation is complete for component i (at the pickup station), and the placement is complete for component $i - n_D$ (at the placement station).

$$tr_i = \max(tp_{i-n_D} + \Delta p,\ tg_i + \Delta g). \qquad i = n_D + 1, \ldots, n_I$$

After the last component has been picked up, n_D rotations are needed to place the components still in the carousel. During this period, the carousel may rotate as soon as the current placement operation has been executed.

$$tr_i = tp_{i-n_D} + \Delta p. \qquad i = n_{I+1}, \ldots, n_{I+n_D}$$

Table movement: tt_i. The table may begin to move as soon as the placement of the previous component is complete.

$$tt_i = tp_{i-1} + \Delta p. \qquad i = 2, \ldots, n_I$$

Magazine movement: tm_i. The magazine rack may begin to move as soon as the placement head has gripped the next component.

$$tm_i = tg_{i-1} + \Delta g. \qquad i = 2, \ldots, n_I$$

By solving the above equations recursively for a given placement sequence, we can determine $t_{jk}(i-1, i)$, and hence $f_i(k)$ for all possible values of k ($k = 1, \ldots, n_M$) and i. For example, to compute $f_2(k)$ it is first necessary to evaluate $t_{jk}(1, 2)$. Substituting into (8) gives:

$$\begin{aligned} t_{jk}(1, 2) &= tg_2 - tg_1 \\ &= tg_2 - 0 = tg_2 \\ &= \max(tr_1 + \Delta r,\ tm_2 + \Delta m_{1,2}) \\ &= \max(tg_1 + \Delta g + \Delta r,\ tg_1 + \Delta m_{1,2}) \\ &= \max(\Delta g + \Delta r,\ tg_1 + \Delta m_{1,2}), \end{aligned}$$

where $\Delta m_{1,2}$ depends on the particular values of k and j used in (7a). The optimal solution is found from: $\min[f_{n_I}(k) : k \in M]$.

The optimal values of x as a function of the state k, come from the solution of (7), and are given by:

$$x_{ij}(k) = \begin{cases} 1 & \text{if } f_i(k) \text{ indicates that component } i \text{ is to be retrieved from magazine position } j; \\ 0 & \text{otherwise.} \end{cases}$$

Each must be stored for use during the recovery operation. It is unlikely, however, that the optimal trajectory will satisfy (1c) given the magazine assignments, y, obtained from the solution to problem (5). Thus, a procedure is needed to find a feasible solution.

4.4. Finding feasible solutions

In this section, we discuss how to obtain an optimal component retrieval sequence, x_{ij}, given the placement sequence and the magazine assignment, y_{jp}. Define a graph $\bar{G}(\bar{V}, \bar{E})$, a subgraph of the one used to solve problem (4). The node set $\bar{V}$ consists of each placement location on the PCB duplicated once for each magazine retrieval position specified by y. That is, if the ith placement requires component type p which is installed in three magazine slots, then there will be three nodes in the layered network associated with the ith placement.

The edge set $\bar{E}$ is formed by connecting the nodes from one placement location to each of the nodes associated with the next placement location. The arc weights are $t_{jk}(i-1, i)$, and once again, must be computed iteratively as the network is traversed.

The corresponding problem is solved using the same dynamic programming scheme described for the quadratic IP. The recursive relation is:

$$f_i(k) = \min[t_{jk}(i-1, i) + f_{i-1}(j)], \qquad i = 2, \ldots, n_I;\ k \in Y_p(i);\ j \in Y_p(i-1) \tag{9}$$

where $Y_p(i)$ is the set of magazine positions containing the component type (call it p) required for the ith placement. The function $f_i(k)$ is interpreted as the minimum time to grip the first i components (and necessarily place the first $i-6$ components) given that the ith component is in magazine position k.

The difference between (7) and (9) is that in the latter, the possible magazine locations from which components can be retrieved is limited by the component assignments obtained from the solution of the transportation problem (5). Also, the term associated with the Lagrange multipliers is absent. Under worst-case conditions, problem (9) has complexity $O(n_I n_M^2)$, with memory requirements $O(n_I n_M)$.

Remark. As mentioned, it is often desirable to stationalize or preassign certain positions on the magazine rack to certain component types. In fact, when all the positions are fixed, problem (9) results. When solving the IP (4) using dynamic program, problem (7) is modified by excluding as a choice for j any preassigned position that does not contain the correct component type. Specifically, if Y_p is the set of preassigned positions for component type p, and component type $\hat{p}$ is in the ith location on the board, then j is restricted as follows: $j \in M \backslash (\cup_{p \neq \hat{p}} Y_p)$. The index k in (7a) is restricted in a similar manner. Finally, when solving the transportation problem (5), the corresponding values of y_{jp} are fixed and the data structures updated accordingly.

5. Input data and experimental design

To evaluate the proposed methodology for a particular board, four sets of data are needed: (1) the *bill of materials*; (2) the *component sequence list* which identifies the component type and its X and Y coordinates; (3) the maximum number of magazine positions per individual component (η_p); and (4) the maximum number of usable slots on the magazine rack.

These data were obtained from Texas Instruments (TI) for 12 different board types currently in production at their Austin plant.[1] For surface mount operations, TI uses the Fuji CP II to place capacitors, resistors, and small outline transistors. Each of these components comes on 8-mm tapes which require only a few seconds for on line setup. Typically, boards contain a limited number of such components that, for the most part, are evenly distributed and densely packed.

Table 2 identifies general characteristics of the 12 board types investigated. The number of placements range from a low of 55 to a high of 500, while the number of different component types on any one board is between 4 and 38. Several statistics are given with respect to the zero penalty table movement distance (20 mm) referred to as Δ. Specific values reflect a combination of placement densities and the operating parameters of the machine. Columns (d) and (e) indicate the average number of component locations on a board that are within one and two Δ, respectively, from each location. In particular, for column (d), this number is computed by determining how many component locations are within one Δ of the ith component location, summing the results over all i, and then dividing the total by n_I, the number of components on a board. So, for Board A, there are on average 6.74 component locations within one Δ, and 23.47 locations within 2Δ, of any (X, Y) coordinate. Columns (f) and (g) give the same statistics for identical components. Because no magazine changes are needed in this case, table movement becomes the limiting factor. Finally, columns (h)

Table 2. Board characteristics.

Board ID	No. of Placements (a)	No. Component Types (b)	Board Size (mm) (c)	No. of All Components Within 1Δ (d)	No. of All Components Within 2Δ (e)	No. of Like Components Within 1Δ (f)	No. of Like Components Within 2Δ (g)	No. of Other Component Types Within 1Δ (h)	No. of Other Component Types Within 2Δ (i)
A	312	4	356 × 257	6.74	23.47	2.13	8.17	1.71	1.99
B	413	16	352 × 257	7.04	27.89	1.44	6.12	2.65	4.46
C	400	8	351 × 254	5.66	23.50	1.12	4.63	2.96	5.38
D	446	16	350 × 256	6.60	26.83	1.78	7.43	2.52	4.40
E	295	5	333 × 233	6.04	18.97	2.41	7.82	1.32	1.64
F	443	9	351 × 251	5.97	30.10	0.85	5.45	3.26	4.98
G	419	8	350 × 246	8.16	26.62	3.37	9.56	1.87	3.00
H	253	4	333 × 211	5.87	21.03	2.51	9.24	1.18	1.53
I	238	25	249 × 155	11.81	37.01	3.39	8.33	3.65	7.68
J	500	23	356 × 255	15.88	58.81	3.86	13.62	5.12	9.14
K	55	16	29 × 28	41.78	54.00	3.60	4.29	13.05	15.00
L	97	38	31 × 50	57.84	93.26	3.26	5.77	26.03	36.12

and (i) indicate the average number of component types within one and two Δ. This gives some insight into the available choices in a typical neighborhood of a placement location.

To date, TI has not been able to obtain good solutions to any of the three subproblems associated with component placement. Currently, they use a naive program provided by the manufacturer to determine the table tour and retrieval sequence, and are investigating a simple heuristic for making the magazine assignments. We compare our approach to a refinement of this heuristic. In either case, problem (9) is solved to find an optimal x given y.

In our modification of TI's heuristic, the maximum number of magazine slots assigned to any component type is based on the cardinality of I_p, and in all instances is limited to 3. If a particular component type is used in less than 10% of the placements, then η_p is restricted to 1. Component types that are used more frequently are allowed a proportionate number of magazine assignments.

With respect to actual magazine positions, assignments are made so that high-volume components are placed together beginning with the center slots, and redundant assignments for a given component type are evenly distributed across the rack. For example, if component types p_1 and p_2 are used for a majority of the placements, they would be assigned adjacent slots whenever possible beginning with positions 50 and 51. This logic reflects current board design. Component types with high placement rates are more likely for a given sequence to be placed consecutively. Also, if a component is assigned to more than one position, its distribution over the magazine should minimize the expected travel time to the nearest slot containing that component.

6. Computational results

All codes developed for the analysis were written in Microsoft Fortran V4.1 and run on an IBM PS/2 Model 80 under DOS 4.0. Solutions to the transportation problem (5) at Step 2 were obtained with a modified verison of Bertsekas's (1985) network flow code. The total number of magazine slots, and the maximum number of slots per component type used in the analysis were provided by the process engineers at TI. Their choices were restricted by the practice of stationalizing 30 slots for high-volume components. Short-range forecasts of demand for each board type motivated the particular values. This input data is given in table 4, column (c), and is seen to vary from board to board.

For sample board E, figure 4 plots the bounds generated by the algorithm for $n_L = 100$ Lagrangian iterations. The lower curve represents values of the relaxed objective function obtained by solving (2). The upper curve plots the objective function value associated with the best feasible point. The difference between the two curves represents the extra time spent moving the magazine carriage to retrieve components. Because it is unlikely that this time will be zero when more than a few distinct components are to be placed, the feasible solutions obtained may be close to optimal. In this particular case, the duality gap is 3.4%.

The results for all 12 boards for 100 iterations are shown in table 3. Column (c) lists the best lower bound generated ($\lambda = 0$). Columns (d) and (e) give the best feasible time (seconds required to place all components), and the iteration at which it is found. The duality gap is shown in column (f). The data indicate that performance drops off in proportion to the number of distinct components to be placed. As a general rule, little improve-

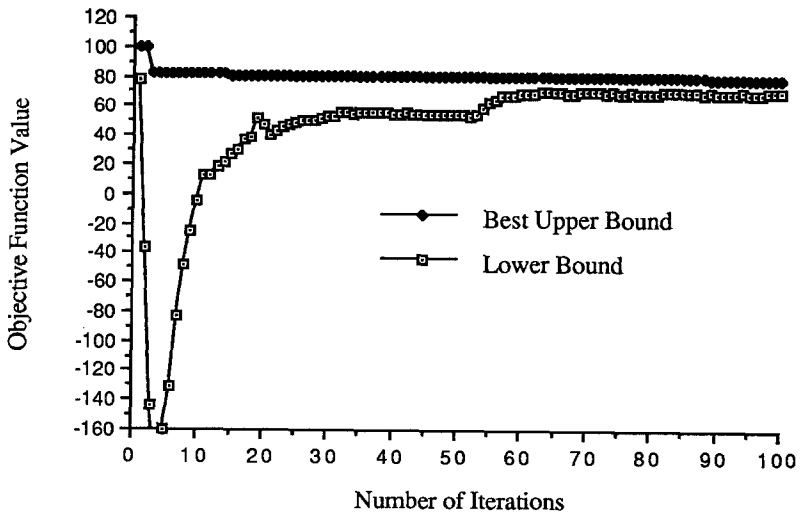


Figure 4. Sample relaxation results for Board E.

Table 3. Results for algorithm and heuristic.

Board ID	No. of Placements (a)	No. Component Types (b)	Lower Bound (c)	Upper Bound (d)	Best Iteration (e)	Duality Gap (%) (f)	Δ Time/ Placements for Relaxation (g)	Heuristic Bound (h)	% Relaxation Improvement (i)
A	312	4	86.18	87.10	39	1.1	0.003	89.29	2.5
B	413	16	107.54	145.94	13	35.7	0.091	195.88	25.5
C	400	8	104.68	142.47	13	36.1	0.095	151.88	6.2
D	446	16	114.84	162.12	99	41.2	0.106	208.28	22.2
E	295	5	77.39	79.99	89	3.4	0.009	83.71	4.4
F	443	9	114.11	158.90	61	39.2	0.101	160.81	1.2
G	419	8	108.10	122.22	45	13.0	0.034	124.60	1.9
H	253	4	64.64	65.35	59	1.1	0.003	66.47	1.7
I	238	25	59.53	117.16	31	96.8	0.243	113.56	−3.2
J	500	23	115.06	159.82	19	38.9	0.090	212.13	24.7
K	55	16	13.50	32.25	11	138.9	0.347	38.85	17.0
L	97	38	24.00	57.15	5	138.1	0.345	44.70	−27.9

ment occurs after 50 iterations for any of the boards; the duality gap stabilizes after about 70 iterations.

Column (g) shows the surplus time above the lower bound averaged over all placements. This represents the average delay per placement caused by extra magazine movement, given the traversal speed of 0.15 seconds per slot.

Results for the heuristic are given in column (h), and compared in percentage terms in column (i) with those obtained with the decomposition procedure. The latter is seen to be superior for all but boards I and L which are distinguished by an extremely low "number

Table 4. Computation times for the decomposition algorithm (seconds).[1]

Board ID	No. of Placements (a)	No. Component Types (b)	No. of Magazines (c)	Placement Sequence (d)	DP (problem 4) (e)	Transportation (problem 5) (f)	Feasible Retrievals (g)	Total Time (h)
A	312	4	12	6.2	3.8	1.5	2.1	7.4
B	413	16	41	10.3	36.9	2.6	3.3	42.8
C	400	8	16	8.5	19.7	2.3	2.7	24.7
D	446	16	45	12.3	36.0	2.6	3.1	41.7
E	295	5	13	4.8	3.7	1.1	0.8	5.6
F	443	9	21	11.1	22.5	1.9	3.3	27.7
G	419	8	24	10.0	21.2	2.0	2.8	26.0
H	253	4	10	3.7	3.8	0.9	2.4	7.1
I	238	25	60	3.9	51.6	3.4	2.0	57.0
J	450	23	23	12.8	69.3	5.8	2.7	77.8
K	55	16	16	0.9	4.0	1.8	2.3	8.1
L	97	38	70	2.0	33.6	3.3	1.8	41.7

[1]Total time per iteration.

of placements" to "number of component types" ratio. The heuristic is ideally suited for such cases because good magazine assignments can be determined by inspection.

Table 4 gives the execution time (I/O + CPU in seconds) per iteration for each component of the relaxation algorithm, as well as the total time consumed by the table tour heuristic. From the data in column (d), it can be seen that the effort required to find a table tour depends directly on the number of placement points. The TSP heuristic is of complexity $O(n_I^2)$.

The amount of time spent solving the DP is shown in column (e), and represents the largest computational burden. Column (f) gives the statistics associated with solving the transportation problem. The algorithm is $O(n_I n_M^2 n_L)$. Column (g) reports the average time spent solving (9) to find a feasible retrieval sequence. Notice that the solution to this problem is obtained in considerably less time than the solution to the parent problem (7), due to the fact that the layered network is now much smaller. Finally, column (h) shows the total time spent solving all three subproblems. It should be noted that a number of improvements in the codes' efficiency are possible by eliminating nonessential I/O. This would produce about a 20% savings for the DP. By way of comparison, the reuslts for the heuristic reported in table 3 were obtained in a slightly greater amount of time than was required to solve the feasible retrieval problem (9).

7. Further analysis

In general, the presence of a duality gap is not uncommon. Darby-Dowman and Lewis (1988) report similar results using a relaxation technique for the capacitated facility location problem, as does Bard (1988) in an analysis of a tool setup problem on a flexible machine (cf. Tang and Denardo 1988). Although Fisher (1981) points to a host of successful applications of Lagrangian relaxation, he warns that such outcomes are highly dependent upon the underlying problem structure.

7.1. Branch and bound

Because the iterative scheme failed to converge satisfactorily in a number of cases, a depth first branch and bound algorithm was developed to test the quality of our solutions. In the implementation, the magazine assignment variables, y_{jp}, are chosen for branching. Recall that if $\bar{y}$ is feasible, the corresponding optimal retrieval sequence, $\bar{x}$, can be found from (9), so in our scheme it is not necessary to branch on x.

Two additional branching rules were adapted to prevent the algorithm from exploring infeasible combinations of magazine assignments. First, for each magazine slot j, it is not permissible for more than a single y_{jp} to be set to 1 on the same path in the tree. Similarly, for each component type p, no more than η_p values of y_{jp} can be set to 1 on the same path. In all cases, a one-hour time limit was placed on the runs.

Even with these modifications, it was difficult to obtain convergence for all board types within the one-hour time limit. This was due to the fact that there are an extremely large number of feasible magazine assignments, and fathoming, based on lower bounds, is not effective in light of Lemma 2. Any partial magazine assignment that does not limit the components that can still be assigned to the remaining magazines will have a lower bound equal to the lower bound derived at the first step in the relaxation. Nevertheless, running the branch and bound code on board types A, E, and H revealed that our decomposition algorithm found the optimum for A, and was within 0.05% of the optimum for the other two.

Finally, we note that reversing the procedure and branching on x_{ij} rather than y_{jp} proved considerably less effective. The difficulty is that almost no fathoming takes place until a depth of n_I is reached in the tree.

7.2. Resequencing the table tour

In an effort to improve upon the above results, a procedure was developed for updating the table tour based on the magazine assignments associated with the best feasible solutions found. The idea is to take advantage of current information and attempt to reach a local optimum. The new sequence is generated as before except now the distance measure used in the TSP heuristic includes time for magazine retrieval. The updated formula is given in Appendix B. Using the new weights produces a reordering of the original sequence that takes into account large magazine movements, and, in general, produces significant reductions in retrieval times. Table 5 gives the results for one iteration of this process. For comparison, the absolute lower bound on total placement time, given that the minimum time between component placements is 0.15 seconds, is listed in column (c). Note that the previously derived lower bounds have no bearing on the new sequence. The best upper bounds from table 3 are repeated in columns (d) and (e) for ease of comparison. The new results for the two models are given in column (f) for the relaxation approach and column (g) for our adaptation of TI's heuristic. They generally show a marked improvement—up to 52.7% and 64.7% in the case of board K [see columns (h) and (i)].

For the most part, the resequencing operation effectively removes costly magazine movements. Nevertheless, a slight degradation in total placement time is observed for boards E, G, and H [see column (h)]. This is attributed to the fact that we simply performed the

Table 5. Results after resequencing.

Board ID	No. of Placements (a)	No. Component Types (b)	Absolute Lower Bound (c)	Upper Bound		Resequenced Upper Bound		% Improvement	
				Relaxation (d)	Heuristic (e)	Relaxation (f)	Heuristic (g)	Relaxation (h)	Heuristic (i)
A	312	4	77.75	87.10	89.29	84.74	85.94	2.7	3.8
B	413	16	103.00	145.94	195.88	121.94	122.82	16.4	37.3
C	400	8	99.75	142.47	151.88	113.55	113.76	20.3	25.5
D	446	16	111.25	162.12	208.29	124.20	132.97	23.4	36.2
E	295	5	73.50	79.99	83.71	80.51	80.72	−0.6	3.6
F	443	9	110.50	158.90	160.81	127.58	128.14	18.9	20.3
G	419	8	104.50	122.22	124.60	124.68	125.76	−2.0	−0.9
H	253	4	63.00	65.35	66.47	67.24	67.23	−2.9	1.1
I	238	25	59.25	117.16	113.56	71.55	72.84	38.9	35.8
J	500	23	112.25	159.82	212.13	122.39	127.52	23.4	39.9
K	55	16	13.50	32.25	38.85	15.24	13.71	52.7	64.7
L	97	38	24.00	57.15	44.70	34.74	24.96	39.2	44.2

resequencing from scratch. In practice, an exchange procedure which only permits adjustments that lead to a reduction in the placement time of an existing sequence would be employed.

8. Discussion and conclusions

The overall results confirm the existence of a strong relationship between the magazine assignment and component retrieval problems, and the need to develop good table tours. We have concentrated on the first two problems at the expense of the latter. More work needs to be done on solving the accompanying TSP. The analysis indicates that performance can be improved greatly by updating the placement sequence in light of the solution found by either the decomposition algorithm or the heuristic. This iterative hierarchical approach has an intuitive appeal and has proven quite useful in practice.

With regard to further research, we note that the development of a procedure for constructing tighter lower bounds would markedly improve the likelihood of convergence during branch and bound. One such approach might involve a closer analysis of consecutive component assignments required by the placement sequence. For example, say component p_1 is placed after component p_2 10 times, after component p_3 20 times, and after component p_4 30 times in the given sequence and that a maximum of 1 magazine position can be assigned for component p_1. We can then deduce that even if component p_1 is assigned to a magazine position adjacent to both components p_3 and p_4, the minimum retrieval time would include at least 10 additional magazine movements to retrieve component p_1 after component p_2. Such information could readily be incorporated in a procedure for generating better lower bounds.

In closing, we suggest an alternative approach to solving the component assignment and retrieval problems. Rather than using subgradient optimization as a mechanism for exploring the feasible region, one might develop heuristics for investigating various magazine assignments and then use the modified layered network algorithm discussed in section 4.4 to gen-

erate feasible solutions. Techniques such as tabu search and simulated annealing have been applied quite successfully in solving related sequencing and assignment problems. The advantage of this approach is that it obviates the need to solve the full quadratic IP (4) at Step 2 of our general algorithm. This is the most time-consuming operation and is required only for updating the Lagrange multipliers. By substituting a heuristic for the transportation problem (5) and using (9) in tandem to find a feasible retrieval sequence, computation times should go down considerably. The quality of the results, however, remains to be seen.

Acknowledgments

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Appendix A

This appendix offers proofs of Lemmas 1 and 2. To begin, let X and Y be the set of constraints associated with the quadratic IP (4) and the transportation subproblem (5), respectively. Define $T(\lambda) \subset X \times Y$ as the set of optimal solutions to the relaxed problem (2) for λ fixed, and note that $T(\lambda)$ is finite. Also, let $(ax - |I|y)$ be shorthand for constraint (1c). With this notation in mind, the following lemma [Theorem 5.23 from Parker and Rardin (1988)] characterizes ascent directions of the dual function $\theta(\lambda)$.

Lemma 3. (Subgradient characterization of Lagrangian dual ascent directions) Let $T(\lambda)$ be the set of optimal solutions to (2). Then $d \in R^{n_M \times n_P}$ is an ascent direction for $\theta(\lambda)$ at a point $\hat{\lambda}$ if and only if

$$d(ax - |I|y) > 0 \qquad \text{for all } (x, y) \in T(\hat{\lambda}).$$

That is, d is an ascent direction if and only if it creates a positive inner product with every subgradient of $\theta(\lambda)$ at $\hat{\lambda}$. For $(\hat{x}, \hat{y}) \in T(\hat{\lambda})$, $a\hat{x} - |I|\hat{y}$ is a subgradient of $\theta(\lambda)$ at $\hat{\lambda}$.

Lemma 4. [Theorem 6.3.1; Bazaraa and Shetty (1979)] $\theta(\lambda)$ is concave for all values of λ.

As a direct consequence of Lemmas 3 and 4, we have:

Corollary. For λ fixed at $\hat{\lambda}$ in (2), let (α^*, d^*) solve the following optimization problem.

$$\max \alpha \tag{A1a}$$

$$\text{s.t.} \quad \alpha \le d(ax - |I|y) \qquad \text{for all } (x, y) \in T(\hat{\lambda}) \tag{A1b}$$

$$d_{jp} \ge 0 \qquad \text{for all } j \text{ and } p \text{ such that } \hat{\lambda}_{jp} = 0 \tag{A1c}$$

$$1 \le d_{jp} \le 1 \qquad \text{for all } j \text{ and } p \tag{A1d}$$

If $\alpha^* = 0$, then $\hat{\lambda}$ solves the Lagrangian dual (3).

Proof of Lemma 1. We will show that for $\lambda = 0$, the solution to problem (A1) is $\alpha^* = 0$. In light of the concavity of $\theta(\lambda)$, this implies that no ascent direction exists at $\lambda = 0$. In the proof, we enumerate a sufficient number of bounding constraints of the form (A1b) to assure that $\alpha \leq 0$. For $\lambda = 0$, note that (A1d) is redundant.

For $\lambda = 0$, one optimal solution to subproblems (4) and (5) is:

$$\hat{x}_{i1} = 1 \qquad i = 1, \ldots, n_I$$

$$\hat{x}_{ij} = 0 \qquad \forall\, j \neq 1;\ i = 1, \ldots, n_I$$

$$\hat{y}_{jp} = 1 \qquad j = p = 1, \ldots, n_P$$

$$\hat{y}_{jp} = 0 \qquad j \neq p,\ j > n_P$$

Here, all components are retrieved from the first magazine slot and each component type is assigned to one of the first n_P magazine positions in ascending order. The value of the objective function $\theta(0)$ is determined by the table tour and the placement sequence, but not the magazine movement.

In the above solution, only the following $n_P - 1$ constraints are violated:

$$\sum_{i=1}^{n_I} a_{ip} x_{i1} \leq |I_p|\, y_{1p} \qquad \forall\, p \neq 1,$$

so we want to include them in (A1b). This gives rise to:

$$\alpha \leq d_{1p} \left(\sum_{i=1}^{n_I} a_{ip} x_{i1} - |I_p|\, y_{1p} \right). \qquad \forall\, p \neq 1 \text{ and } (x, y) \in T(0) \tag{A2}$$

For the remaining constraints in (1c) which are satisfied at $(\hat{x}, \hat{y})$, it is desirable to set $d_{jp} = 0$.

Consequently, the only possible ascent directions must be in the set $\{d : d_{1p} \geq 0$ for $p \neq 1$, and $d_{jp} = 0$ for $j \neq 1\}$. Now consider a second optimal solution to (A1):

$$\bar{x}_{i2} = 1 \qquad i = 1, \ldots, n_I$$

$$\bar{x}_{ij} = 0 \qquad \forall\, j \neq 2,\ i = 1, \ldots, n_I$$

$$\bar{y}_{jp} = \hat{y}_{jp} \qquad \forall\, j \text{ and } p.$$

Here, all components are retrieved from the second magazine slot. For $j = 1$ and $p \neq 1$ we have:

$$\bar{x}_{i1} = 0 \qquad i = 1, \ldots, n_I$$

$$\bar{y}_{1p} = 0 \qquad p \neq 1.$$

Substituting these values into (A2) gives

$$\alpha \leq d_{1p}\left(\sum_{i=1}^{n_I} a_{ip}\,\bar{x}_{i1} - |I_p|\,\bar{y}_{1p}\right) = 0. \qquad \forall\, p \neq 1.$$

Thus, $\alpha \leq 0$ so from the Corollary we can conclude that no ascent direction exists at $\lambda = 0$. ■

Proof of Lemma 2. It can be seen from the preceding that the solution $\theta(0)$ represents a zero penalty schedule. That is, all components are picked from the same slot so no time is lost waiting for the magazine rack to reach the pickup station. Because $\lambda = 0$ is the optimal solution to the dual problem, it follows that any feasible solution differs from $\theta(0)$ by the additional delay introduced by the feasible component retrieval sequence x. ■

Appendix B

The *table tour* problem can be thought of as a TSP where each location on the board must be visited once, and where the distance between two locations is a function of the time required to move the table between them. The objective is to minimize the total travel time.

There are many good heuristics for solving the TSP (Lawler et al. 1986). None of them, however, has been designed with the notion of free movement in mind. For our problem, the minimum time tour may not be the best from the standpoint of maximizing board throughput. In any case, our primary interest in this paper is to solve the *component assignment* and *component retrieval* problems. Thus, we have chosen a simple heuristic based on the weighted distance to the nearest neighbor. The formula used to compute this distance is given by:

$$c_{ij} = w_{ij} \max[\delta X \cdot |X_j - X_i|,\ \delta Y \cdot |Y_j - Y_i|], \tag{B1}$$

where c_{ij} is the weighted distance between placement position i and j; w_{ij} is the weighting factor; X_i, Y_i are the coordinates of location i on the PCB; and δX, δY are the X and Y table movement rates.

The heuristic starts at reference point (0, 0) and visits the closest unvisited neighbor until the tour is complete. The weighting factor w_{ij} is assigned a value of 1 if locations i and j require the same component, and 1.15 otherwise. The latter value is intended to account for the need to move the magazine at least one slot. Of course, the number of magazine movements is unknown, and even if identical components are placed sequentially, it might be desirable to pull them from different locations. Therefore, this weighting scheme is only an approximation.

After a sequence is determined, postprocessing is performed to improve the results. The table movement times are examined and a 2-exchange procedure is called to shorten the overall time of the tour. Results are contained in Clayton (1989).

Resequencing the tour

Once problem (1) is solved and a feasible magazine assignment is available, we generate a new sequence using this information. In particular, a second term is added to the right-hand side of (B1) to account for magazine movement between successive placements. The updated formula is

$$c_{ij} = w_{ij} \max[\delta X \cdot |X_j - X_i|, \delta Y \cdot |Y_j - Y_i|\} + \max\{\Delta r, \delta m \cdot |m_i - m_j|], \quad \text{(B2)}$$

where δm is the magazine movement rate (see table 1); m_i is the magazine slot from which the ith component is retrieved; and m_j is the nearest feasible magazine from which to retrieve the jth component. After the ith placement, the cost of moving to any neighbor j is increased by the amount $\max\{\Delta r, \delta m \cdot |m_i - m_j|\}$. This is intended to compensate for the magazine movement.

Appendix C

Table C1. Data for board K.

Sequence Number	Position		Part Type Number	Rotation (degrees)	Part Name	TI Identification Number
	X	Y				
1	−3048	3587	1	0′	Q9	1095481-0001
2	−2286	1778	2	180′	R8	1113437-0001
3	−2286	2000	3	180′	R7	1101829-0001
4	−3906	3683	4	270′	CR2	1113370-0001
5	−4001	2857	4	0′	CR1	1113370-0001
6	−2794	1301	5	90′	R21	1113436-0001
7	−3461	1301	5	90′	R5	1113436-0001
8	−3017	1301	6	90′	R20	1103061-0001
9	−3747	1428	6	180′	R4	1103061-0001
10	−4699	4127	7	90′	R36	1113431-0001
11	−4922	2571	7	180′	R34	1113431-0001
12	−4445	4127	8	90′	R35	1103067-0001
13	−4922	2317	8	180′	R33	1103067-0001
14	−2540	2952	9	180′	C2	1103281-0001
15	−3334	2317	9	90′	C1	1103281-0001
16	−2255	3175	10	180′	C15	1101810-0001
17	−3334	2825	10	90′	C14	1101810-0001
18	−2159	2698	11	90′	R1	1113438-0001
19	−2255	3429	11	180′	R39	1113438-0001
20	−2255	1555	11	180′	R38	1113438-0001
21	−2636	1682	12	270′	Q10	1095460-0001
22	−2985	3206	12	0′	Q8	1095460-0001
23	−3144	1682	12	270′	Q7	1095460-0001
24	−3715	2254	12	270′	Q6	1095460-0001
25	−4160	1905	12	270′	Q5	1095460-0001
26	−2159	2317	13	90′	R44	1101732-0001
27	−2572	1301	13	90′	R22	1101732-0001

Table C1. Continued.

Sequence Number	Position X	Position Y	Part Type Number	Rotation (degrees)	Part Name	TI Identification Number
28	−2921	3905	13	180′	R103	1101732-0001
29	−3239	1301	13	90′	R6	1101732-0001
30	−3366	3143	13	180′	R104	1101732-0001
31	−3747	3143	13	180′	R101	1101732-0001
32	−4604	1968	13	180′	R102	1101732-0001
33	−2413	4127	14	180′	C9	1102992-0001
34	−4128	1397	14	180′	C6	1102992-0001
35	−4160	4000	14	90′	C8	1102992-0001
36	−4509	2508	14	90′	C11	1102992-0001
37	−4953	4127	14	90′	C13	1102992-0001
38	−4953	1555	14	180′	C10	1102992-0001
39	−5080	1873	14	90′	C12	1102992-0001
40	−2699	3683	15	180′	R28	1126225-0001
41	−2921	2889	15	180′	R19	1126225-0001
43	−3525	3841	15	180′	R29	1126225-0001
43	−3525	3587	15	180′	R30	1126225-0001
44	−3842	2540	15	180′	R15	1126225-0001
45	−4033	2254	15	90′	R10	1126225-0001
46	−4128	1619	15	180′	R3	1126225-0001
47	−4255	2508	15	90′	R16	1126225-0001
48	−2921	4127	16	180′	R25	1126229-0001
49	−3239	3873	16	90′	R26	1126229-0001
50	−3334	3365	16	180′	R24	1126229-0001
51	−3556	2571	16	90′	R14	1126229-0001
52	−3620	2889	16	180′	R13	1126229-0001
53	−3715	3365	16	180′	R23	1126229-0001
54	−4350	2190	16	180′	R12	1126229-0001
55	−4604	1746	16	180′	R11	1126229-0001

Note

1. The full set of data for all the boards can be obtained from the first author.

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